Section 1.5 — Linear Functions — Answer Key

Math 103: College Algebra

Identifying Linear Expressions, Equations, and Functions

Identify each expression as linear or not linear. If the expression is not linear, mark any parts of the expression that make it not linear.

1. Linear
2. $\dfrac{3x-2}{4x+1}$
3. $x^2 + 2x + 5$
4. Linear

Identify each equation as linear or not linear. If the equation is not linear, mark any parts of the equation that make it not linear.

5. $(5x+2)(3x-1) = 14$
6. $\sqrt{x+2} = x - 4$
7. Linear
8. Linear

Identify each function as linear or not linear. If the function is not linear, mark any parts of the function that make it not linear.

9. Linear
10. $f(x) = 3x + x^3$
11. $f(x) = |4x - 3|$
12. Linear

Writing Linear Functions from Graphs

For each function graphed below:

a. Write an equation for the function in point-slope form, using a point other than the y-intercept. Use f as the name of the function.

b. Simplify your answer to give the function in slope-intercept form.

c. Give the coordinates of the line's y-intercept. Use your equation from part (b) if it is not clear on the graph.

13. a. $f(x) = -\frac{4}{3}(x-3) - 2$
    b. $f(x) = -\frac{4}{3}x + 2$
    c. $(0, 2)$

14. a. One of two equations:
    $f(x) = \frac{3}{5}(x+3) + 1$
    $f(x) = \frac{3}{5}(x-2) + 4$
    b. $f(x) = \frac{3}{5}x + \frac{14}{5}$
    c. $(0, \frac{14}{5})$

Graphing Linear Functions

For each function below:

a. Fill in the table to show three points on the graph with integer coordinates.

b. Use two of your points to graph the function.

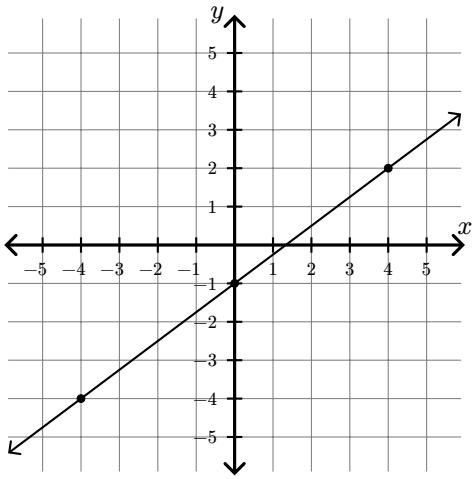
c. Confirm that the graph passes through the third point.

Note: If you were using the technique shown in the videos, you should have exactly the three points shown in the answer key.

15.

$f(x) = \frac{3}{4}x - 1$

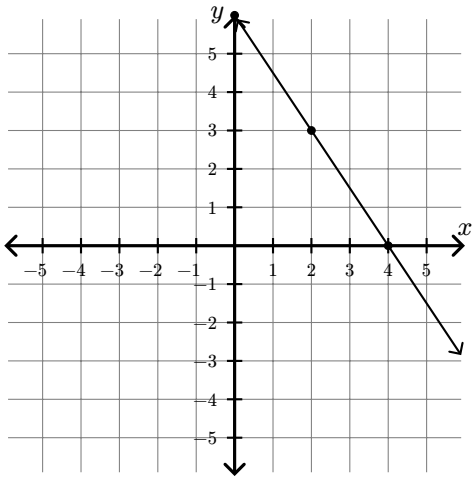
x	$y = f(x)$
-4	-4
0	-1
4	2



16.

$f(x) = -\frac{3}{2}(x - 2) + 3$

x	$y = f(x)$
0	6
2	3
4	0



Inverses of Linear Functions

Find the inverse of each linear function below. Show your work carefully on a separate sheet of paper.

17. $f^{-1}(x) = \frac{1}{2}x + 4$

18. $g^{-1}(x) = \frac{7}{2}x - 21$

19. $h^{-1}(x) = -\frac{3}{4}x + \frac{3}{2}$

20. $k^{-1}(x) = -\frac{1}{3}x - \frac{5}{3}$

21. $p^{-1}(x) = 2x - 6$

22. $q^{-1}(x) = \frac{9}{7}x - \frac{162}{7}$